

MANISH PATHAK

Embedded Systems Developer • Aerospace Enthusiast • Student

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PERSONAL PROFILE

Self-taught embedded systems developer and student with hands-on experience in microcontrollers, wireless telemetry, IoT devices, and hardware prototyping. Skilled in programming platforms such as ESP32, ESP8266, and Raspberry Pi Pico using C and Python. Practical knowledge of custom sensor integration, wireless communication protocols, and basic cybersecurity testing. Focused on practical hands-on engineering, breaking down complex systems, and building long-term initiatives in aerospace and technology.

TECHNICAL SKILLS

Embedded Systems & Hardware:	ESP32, ESP8266, Raspberry Pi Pico, Arduino, Telemetry Modules (NRF24L01), Barometric & Motion Sensors (BMP180, ADXL345, Ultrasonic), Servos, Relays, Circuit Prototyping.
Programming & Software:	C, Python, JavaScript, HTML5, CSS3, WebSockets, Blynk IoT Platform.
Cybersecurity & Tools:	Kali Linux, Network Pentesting, Nmap, Burp Suite, Hydra, Hashcat.
Prototyping & Concepts:	Circuit assembly, telemetry data logging, wireless communication protocols, 3D printing & airframe structural design.

PRACTICAL PROJECTS

Model Rocket & Telemetry Flight Computer

Aerospace & Avionics Prototype

- Built a flight computer prototype around an ESP32 microcontroller to record and transmit flight telemetry during launch.
- Integrated a BMP180 barometric sensor for altitude estimation, an ADXL345 accelerometer, and microSD card data logging.
- Designed a radio telemetry link using NRF24L01 modules and added a servo-actuated parachute deployment system.

SU-27 Remote-Controlled Aircraft with Telemetry

Embedded Control & Flight Avionics

- Developing an RC aircraft modeled after the Sukhoi Su-27 using a Raspberry Pi Pico flight controller, brushless motor, and ESC driver.
- Built custom telemetry streaming for altitude and sensor data using NRF24L01 wireless transceiver modules.
- Created a custom WebSocket software ground interface to view flight telemetry and send control commands directly from a PC.

Smart Assistive Glasses for Visually Impaired Users

Assistive Technology & Microcontrollers

- Built a wearable assistive device using an ESP8266 microcontroller to help visually impaired individuals detect nearby obstacles.
- Integrated ultrasonic distance sensors to trigger real-time audio buzzer alerts whenever an object enters a safety margin.

Wi-Fi Controlled Robotic Car

Robotics & IoT Systems

- Designed and programmed a 4-wheel robotic vehicle using an ESP32 microcontroller controlled wirelessly over Wi-Fi.
- Integrated real-time movement commands via the Blynk mobile application to drive motors with minimal latency.

Smart Home Automation System

IoT & Electrical Control

- Created a wireless home control setup using an ESP8266 module paired with a 2-channel relay to switch household appliances remotely.

Mini Solid-State Tesla Coil

Electromagnetics & Circuit Design

- Constructed a high-frequency solid-state Tesla coil circuit using a 2N2222A transistor to demonstrate wireless energy transmission.
- Successfully generated high-frequency electric fields capable of wirelessly illuminating CFL bulbs.

EDUCATION & CERTIFICATIONS

- **Advent of Cyber 2025 Certificate** — TryHackMe (*Hands-on training in network security, web security, and pentesting tools*)
- **Secondary Education Examination (SEE / Class 10)** — Marksheet Credentials
- **Self-Directed Engineering & Computer Science Learning** — Continuous self-study in embedded systems, C programming, Linux administration, and aerospace avionics.

INTERESTS & ACTIVITIES

- **Aerospace & Space Technology:** Working on community and research initiatives (Rise Nepal) to support aerospace learning and space technology in Nepal.
- **Technical Reading:** Studying documentation, circuit manuals, and aerospace engineering books.
- **Drawing & Concept Sketching:** Creating technical drawings and structural concept sketches for hardware projects.